Grade 6 Accelerated
Unit 10 Assessment

Name____________________
Date__________________
Period__________

Showing Work with a Calculator: Because you are using a calculator, it is possible to accidently punch the wrong button. To ensure you earn partial credit, in each work box show the operation(s) that you entered into the calculator and the result.

1. A real estate agent tracked the prices of homes sold in a specific neighborhood. The stem-and-leaf plot of the home prices, in thousands, for the homes sold in the neighborhood is shown.

Key	$18|3 = \$183{,}000$

Home Price

Stem	**Leaf**
9	1
10	
11	
12	
13	
14	
15	
16	
17	1, 2, 5, 8
18	0, 3, 7
19	2, 4

Complete the statement. Show your work.

a. The measure of center that should be used to represent the typical sale price of the homes in the neighborhood is the

- ○ mean
- ○ median

home price. The typical price for the neighborhood is $179,000.

work

b. The data

- ○ has an outlier,
- ○ is generally symmetric,

so

- ○ the range is
- ○ the interquartile range is
- ○ both the range and the interquartile range are

the preferred measure(s) of variation.

2. The annual lightning report gives the lightning density, the number of lightning strikes per square kilometer (strikes per km^2), for each US state. A researcher uses the information in the report to compare the lightning strikes in the southern US states and in the western US states. The researcher's data is shown.

Southern US States		**Western US States**	
State	**Lightning Density (strikes/km^2)**	**State**	**Lightning Density (strikes/km^2)**
Alabama	37.87	**Arizona**	12.10
Arkansas	41.09	**California**	1.03
Florida	85.99	**Colorado**	12.80
Georgia	35.95	**Montana**	5.19
Louisiana	81.62	**Nevada**	2.55
Mississippi	55.79	**New Mexico**	16.24
Missouri	42.60	**Oregon**	0.77
Oklahoma	54.91	**Utah**	5.72
Tennessee	35.87	**Washington**	0.30
Texas	60.25	**Wyoming**	6.17

Complete each statement. Show your work. If necessary, round to the nearest hundredth.

a. The southern states have a median of 48.76 strikes/km^2 and a

mean of 53.19 strikes/km^2.

work

b. The western states have a median of 5.46 strikes/km^2 and

a mean of 6.29 strikes/km^2.

work

c. Neither data set has an outlier. Therefore, the
- mean
- median

should represent the center and the
- range
- interquartile range

should be used to represent variability.

d. The western states tend to have
- less
- more

variability than the

southern states because the
- range
- interquartile range
- range and the interquartile range

is/are
- larger
- smaller

than the
- range
- interquartile range
- range and the interquartile range

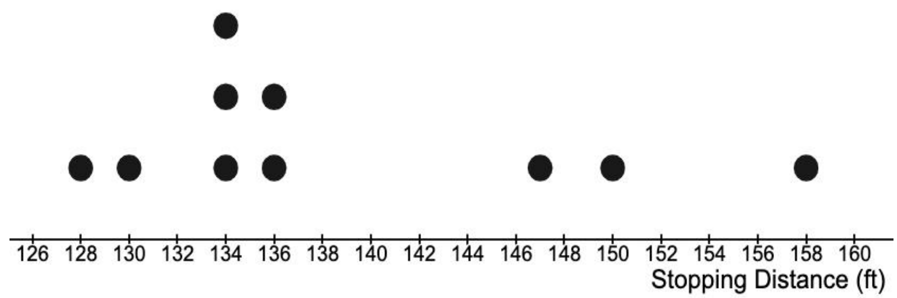
for the southern states.

3. A tire manufacturer is testing the braking performance of one of its tire models on a test track. The company evaluated the tires on 10 different cars, recording the stopping distance, in feet (ft), for each car driving 50 miles per hour on both wet and dry pavement. The data is shown.

Stopping Distance on Wet Pavement

Car	Stopping Distance on Wet Pavement (ft)
A	201
B	220
C	192
D	146
E	182
F	173
G	202
H	180
K	192
L	206

Stopping Distance on Dry Pavement



Complete the statements.

a. A car's stopping distance on dry pavement is typically
- ○ less
- ○ more

than the stopping distance on wet pavement. The stopping distance on dry pavement has
- ○ less
- ○ more

variability than the stopping distance on wet pavement.

b. The stopping distance of car *D* on wet pavement is
- ○ a gap
- ○ a cluster
- ○ an outlier

which means that the
- ○ mean
- ○ median

is the best measure of center and the
- ○ range
- ○ interquartile range

is the best measure of spread for this data.

4. Terri recorded her resting pulse rate at different times of the day. Her data is shown and is labeled as data set A.

$$A = \{70,70,70,70,72,72,72,73,76,76,77,77,78,78,78,80,80,80\}$$

Terri added two measurements of her pulse rate during exercise to create the new data set B, as shown.

$$B = \{70,70,70,70,72,72,72,73,76,76,77,77,78,78,78,80,80,80,120,142\}$$

The mean of data set A is $74.9\overline{4}$. The mean of data set B is 80.55.

Complete the statement.

The measure of center that should be used for data set B is the

- ○ mean
- ○ median

because the two additional pulse rates that were added to data set A to create data set B influenced the

- ○ range.
- ○ interquartile range.

5. After a new process of producing a computer chip was introduced at a manufacturing plant, a random sample of 640 of the computer chips showed that 8 of the chips had a defect.
Determine the number of defective computer chips when $100{,}000$ computer chips are manufactured.

work	
Defective Computer Chips	1250

6. Talking or texting on a cell phone can be a distraction. A survey of 80 middle schoolers at Sebastian Middle School found that 60% of those surveyed had someone who was talking or texting walk into them. Another survey conducted by a cell phone company found that of 738 of cell phone owners, 23% of those surveyed had someone who was talking or texting walk into them.

a. What are the two populations that were surveyed?

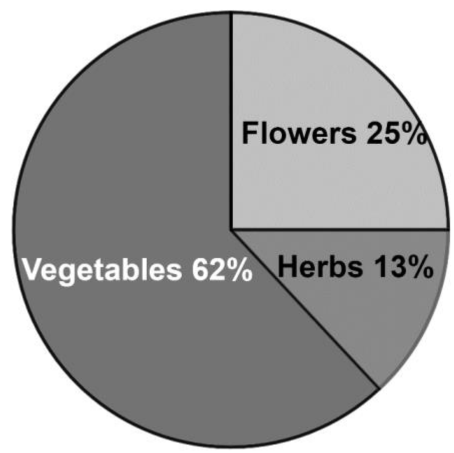
Populations	Middle schoolers at Sebastian Middle School and cell phone owners

b. Complete the statement.

- ○ Both surveys
- ○ Neither survey
- ○ The survey conducted by a cell phone company
- ○ The survey conducted at Sebastian Middle School

would be a representative sample of all cell phone users.

7. In 2015, a seed company surveyed people to ask them what type of seeds they plan to buy. The results are shown.



The population of Yulee, Florida in 2015 was $11{,}672$. Use the results to predict the number of people in Yulee who may have bought flower seeds in 2015.

work	
People in Yulee Who May Have Bought Flower Seeds	2918